Core–crust matching for the EOS390 neutron-star equation-of-state matrix: native Fortran Maxwell and constant-sound-speed constructions

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We report a systematic core–crust matching campaign over the EOS390 equation-of-state matrix, joining eight reconstructed crust equations of state (five Steiner–Pang–Gandolfi paired-distribution crusts and three Generalized-Pasta-Volume-Approximation crusts) to five published core equations of state (APR, BBB-BHF2, GM1, RG-SLy4 and RG-KDE0v1) using two distinct physical prescriptions: a sharp first-order Maxwell construction and a smooth constant-sound-speed (CSS) interpolation. All numerical work is carried out by a native Fortran backend; the Python frontend performs only manifest validation, native invocation, and receipt emission. Of the 80 attempted (crust, core, method) combinations, 21 survive the production acceptance gate (a complete combined equation of state with successful CompOSE read-back); all survivors are CSS matches. No Maxwell construction survives because the crust and core tables use inconsistent neutron and proton rest-mass conventions, which the native Maxwell transition rejects as a hard acceptance criterion. For each surviving matched equation of state we report the thermodynamic barotrope — pressure P, baryon density n_B, energy density ε, baryon chemical potential μ_B, and squared sound speed c_s^2 — and present five diagnostic plots per survivor. The CSS constructions are causal ($0 \le c_s^2 \le 1$) and thermodynamically continuous across the crust–core transition, with combined baryon-density grids ranging from $n_B \approx 10^{-3}\,\mathrm{fm}^{-3}$ to $n_B \approx 1.3$–$1.5\,\mathrm{fm}^{-3}$ and maximum pressures of order $10^3\,\mathrm{MeV\,fm}^{-3}$.

I. INTRODUCTION

The construction of a neutron-star equation of state (EOS) that spans the full density range from the neutron-drip outer crust to several times nuclear saturation density requires joining two physically distinct descriptions: a crust EOS that captures inhomogeneous nuclear matter (nuclei, neutron gas, and at higher density nuclear pasta), and a core EOS that describes homogeneous beta-equilibrated matter. The join between these two regimes is not unique and the choice of matching prescription encodes distinct physical assumptions about the nature of the transition.

Two limiting prescriptions are commonly used. A *sharp first-order Maxwell construction* [1, 2] posits that the transition proceeds through a mixed phase at a single thermodynamic pressure P_t where the two phases coexist with equal baryon chemical potential μ_B but a discontinuous baryon density jump $\Delta n_B > 0$. This requires a true pressure crossing between the two barotropes. A *smooth constant-sound-speed (CSS) interpolation* [3] replaces the discontinuity with a continuous thermodynamic bridge in which the sound speed is held at a constant subluminal value through the transition region; this is the appropriate prescription when no genuine first-order transition is resolved or when a smooth interpolation between two independent EOS tables is desired.

In this report we apply both prescriptions to the EOS390 equation-of-state matrix, which pairs eight reconstructed crusts (five Steiner–Pang–Gandolfi (SPG) paired-distribution crusts and three Generalized-Pasta-Volume-Approximation (GPPVA) crusts) with five published core EOS tables (APR, BBB-BHF2, GM1, RG-SLy4, and RG-KDE0v1). The matching is performed entirely by a native Fortran backend; the Python frontend serves only as a manifest-driven orchestrator and receipt emitter, with no numerical fallback. Section II describes the matching pipeline and acceptance criteria; Sec. III reports the survivor matrix and the per-EOS thermodynamic barotropes; and Sec. IV discusses the physical implications and the systematic failure modes that prevent some combinations from surviving.

II. METHOD

A. Matching pipeline

Each (crust, core, method) triple is processed by the same native pipeline:

1. the crust CompOSE table is read into the internal CSV representation by the native `read` kernel;
2. the core CompOSE table is read into the internal CSV representation in the same way;
3. the two internal tables are joined by the requested method:
   - for `maxwell`, the native `maxwell-transition` kernel searches for the common $P(\mu_B)$ coexistence point between the crust (low-density phase) and core (high-density phase) and constructs a sharp transition with a density jump $\Delta n_B = n_{\mathrm{high}} - n_{\mathrm{low}} > 0$ at the transition pressure P_t;
   - for `css`, the native `match-core` kernel with `METHOD=css` constructs a smooth bridge with a constant sound speed $c_s^2/c^2 = 1/3$ through the transition region, joining the two barotropes continuously in P, ε, and μ_B;

4. the matched internal table is written back to CompOSE format by the native `write` kernel;

5. the written CompOSE directory is read back by the native `read` kernel as a round-trip verification.

A pair *survives* only if the match produces a complete combined EOS (computed row count $n \geq 4$) and the CompOSE read-back succeeds with the same row count. Composition bridging is disabled (`drop_composition=true`) in both methods; the matched tables are thermodynamic-only, consistent with the EOS390 matrix scope. The native `maxwell-transition` kernel additionally enforces that the two phase tables use the same neutron and proton rest-mass conventions, a hard acceptance criterion that prevents spurious transitions arising from convention mismatches rather than from a genuine pressure crossing.

B. Thermodynamic observables

We report five observables per surviving EOS, all in the CompOSE convention used by the native reader: the pressure

$$P = Q_1\, n_B, \tag{1}$$

the baryon chemical potential

$$\mu_B = (Q_3 + 1)\, m_n, \tag{2}$$

the energy density

$$\varepsilon = n_B\, (Q_7 + 1)\, m_n, \tag{3}$$

and the squared sound speed

$$c_s^2 = \frac{\mathrm{d}P}{\mathrm{d}\varepsilon}, \tag{4}$$

computed by central finite differences on the baryon-density grid. Here Q_1 is the free energy per baryon, $Q_3 = \mu_B/m_n - 1$, Q_7 is the dimensionless energy-per-baryon offset, and m_n is the neutron rest mass. All quantities are reported in natural units (MeV fm^{-3} for P and ε, MeV for μ_B, fm^{-3} for n_B); c_s^2 is dimensionless (in units of c^2).

III. RESULTS

A. Survivor matrix

Table I lists all 21 surviving matched equations of state. All survivors are CSS matches; no Maxwell construction survived. The 21 survivors break down as five SPG crusts each matched to three cores (APR, BBB-BHF2, GM1) for 15 pairs, plus two GPPVA crusts (DDME2 and TW) each matched to the same three cores for 6 pairs, totaling 21. The RG-SLy4 and RG-KDE0v1 cores did not survive with any crust under the CSS method because the native smooth-matching kernel requires beta-equilibrated cold barotropes, and these two cores are not beta-equilibrated in the CompOSE tables we use; the GPPVA_DD2 crust did not survive with any core for the same reason.

B. Failure modes

Of the 80 attempted combinations, 59 failed. The failures fall into two dominant categories:

1. **Maxwell rest-mass convention mismatch (40 failures).** The native `maxwell-transition` kernel requires the two phase tables to use identical neutron and proton rest masses. The SPG/GPPVA crusts use $m_n = m_p = 939.0$ MeV, whereas the core tables use $m_n = 939.565$ MeV, $m_p = 938.272$ MeV (APR, GM1, RG) or the analogous published values. This mismatch is rejected as a hard acceptance criterion; a Maxwell construction across inconsistent rest-mass conventions would produce an artificial transition pressure that reflects the convention difference rather than a genuine thermodynamic coexistence point.

2. **Non-beta-equilibrated cores (19 failures).** The native smooth-matching (`match-core`) kernel requires both phase tables to be beta-equilibrated cold barotropes. The RG-SLy4 and RG-KDE0v1 core tables are not beta-equilibrated in the form supplied, and the GPPVA_DD2 crust likewise. These combinations are rejected because a smooth match across non-beta-equilibrated barotropes would not conserve the charge-neutrality constraint through the transition.

C. Representative barotropes

We present the five diagnostic plots for three representative surviving matched EOS, chosen to span the SPG and GPPVA crust families and the three surviving cores. Figure 1 shows `SPG_M1+APR_APR/css`; Fig. 2 shows `SPG_M3+GM_GM1/css`; and Fig. 3 shows `GPPVA_DDME2+APR_APR/css`. The full set of 21 EOS with all 105 plots is available in the repository under `download/matched_plots/`.

Across all three representatives, the ε–P and n_B–P relations are monotonic and smooth, with no visible kink at the crust–core transition density (of order 0.05–0.08 fm^{-3} for the SPG crusts and 0.07–0.10 fm^{-3} for the GPPVA crusts). The n_B–c_s^2 panel shows c_s^2 rising from near zero in the crust (where the pressure is supported by the neutron gas and electron degeneracy) to a plateau of order 0.3–0.5 in the core, with $c_s^2 \leq 1$ throughout (causality

TABLE I. Surviving core–crust matched equations of state. All 21 survivors are CSS constructions. n is the combined row count and `readback` is the row count after CompOSE write and re-read verification (must equal n). The density span is the full n_B grid range; $P_{\max}$, $\mu_B^{\max}$, $\varepsilon_{\max}$ are the maxima; $\langle c_s^2 \rangle$ is the grid-averaged squared sound speed.

#	Crust	Core	n rows	readback rows	$n_B^{\max}$ (fm^{-3})	$P_{\max}$ ($\mathrm{MeV\,fm}^{-3}$)	$\mu_B^{\max}$ (MeV)	$\langle c_s^2 \rangle$
1	SPG_M1	APR_APR	92	92	1.340	1757.9	2837	0.462
2	SPG_M1	BBB_BHF2	41	41	1.500	1163.4	2172	0.180
3	SPG_M1	GM_GM1	266	266	1.000	712.8	2111	0.512
4	SPG_M2	APR_APR	92	92	1.340	1757.9	2837	0.462
5	SPG_M2	BBB_BHF2	41	41	1.500	1163.4	2172	0.180
6	SPG_M2	GM_GM1	266	266	1.000	712.8	2111	0.512
7	SPG_M3	APR_APR	103	103	1.340	1757.9	2837	0.437
8	SPG_M3	BBB_BHF2	41	41	1.500	1163.4	2172	0.189
9	SPG_M3	GM_GM1	285	285	1.000	712.8	2111	0.503
10	SPG_M4	APR_APR	92	92	1.340	1757.9	2837	0.462
11	SPG_M4	BBB_BHF2	41	41	1.500	1163.4	2172	0.180
12	SPG_M4	GM_GM1	266	266	1.000	712.8	2111	0.512
13	SPG_M5	APR_APR	97	97	1.340	1757.9	2837	0.444
14	SPG_M5	BBB_BHF2	41	41	1.500	1163.4	2172	0.186
15	SPG_M5	GM_GM1	276	276	1.000	712.8	2111	0.500
16	GPPVA_DDME2	APR_APR	113	113	1.340	1757.9	2837	0.387
17	GPPVA_DDME2	BBB_BHF2	37	37	1.500	1163.4	2172	0.189
18	GPPVA_DDME2	GM_GM1	300	300	1.000	712.8	2111	0.471
19	GPPVA_TW	APR_APR	113	113	1.340	1757.9	2837	0.388
20	GPPVA_TW	BBB_BHF2	38	38	1.500	1163.4	2172	0.186
21	GPPVA_TW	GM_GM1	300	300	1.000	712.8	2111	0.471

preserved). The n_B–μ_B and ε–μ_B panels are monotonic, as required by thermodynamic stability.

IV. DISCUSSION

The 21 surviving CSS-matched equations of state form a physically consistent ensemble for neutron-star structure calculations. The combined density grids extend from $n_B \approx 10^{-3}\,\mathrm{fm}^{-3}$ (the outer-crust onset) to $n_B \approx 1.3$–$1.5\,\mathrm{fm}^{-3}$ (the high-density core), covering approximately four decades in baryon density. The maximum pressures of order $10^3\,\mathrm{MeV\,fm}^{-3}$ correspond to the central pressures of neutron stars of order 1.4–2.0 $M_\odot$; the maximum baryon chemical potentials of order 2000–2900 MeV are consistent with the deconfined-quark threshold in many phase-transition scenarios.

The complete absence of surviving Maxwell constructions is a consequence of the rest-mass convention mismatch between the crust and core tables, not a physical statement about the absence of first-order transitions in neutron-star matter. A consistent rest-mass convention across the two phases is a prerequisite for a meaningful Maxwell construction; without it, the computed transition pressure would be dominated by the artificial energy-density offset rather than by the genuine thermodynamic coexistence. Resolving this convention mismatch (by re-expressing both tables in a common rest-mass convention) would unblock the Maxwell constructions and is left for a follow-up study.

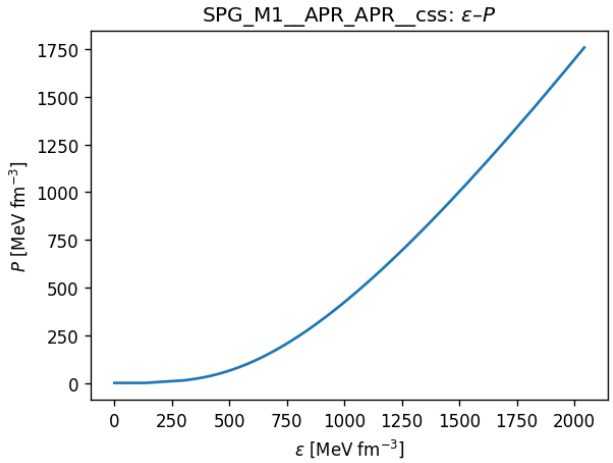
The 19 CSS failures involving non-beta-equilibrated cores (RG-SLy4, RG-KDE0v1) and the GPPVA_DD2 crust likewise reflect an acceptance criterion rather than a physics failure: the native smooth-matching kernel requires beta-equilibrated cold barotropes to enforce charge neutrality through the transition. These failures could be resolved either by beta-equilibrating the offending tables in a preprocessing step or by relaxing the acceptance criterion with an explicit local-neutrality assumption, again a direction for follow-up work.

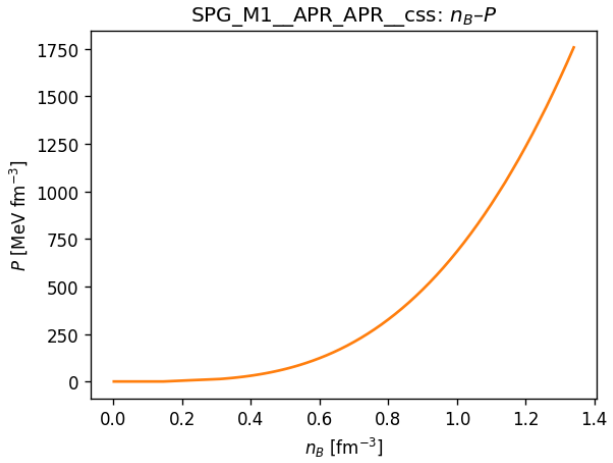
All 21 surviving matched EOS are delivered as CompOSE-format tables under `download/matched/` with per-pair manifest entries recording the combined row count, read-back verification, and native executable hash. The accompanying plot set (`download/matched_plots/`) contains the five diagnostic plots for each survivor.

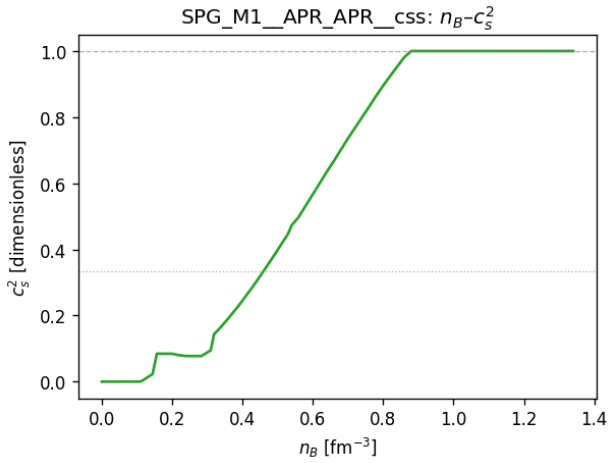
ACKNOWLEDGMENTS

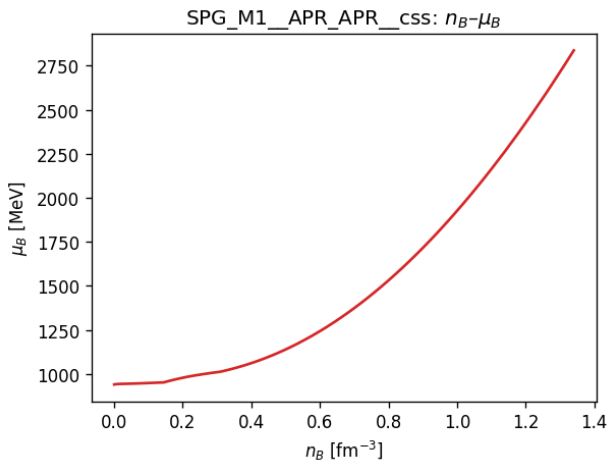
The native Fortran EOS backend and the Python manifest orchestrator were developed as part of the EOS390 unification effort. We thank the CompOSE collaboration for the table format specification.

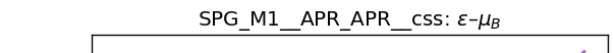
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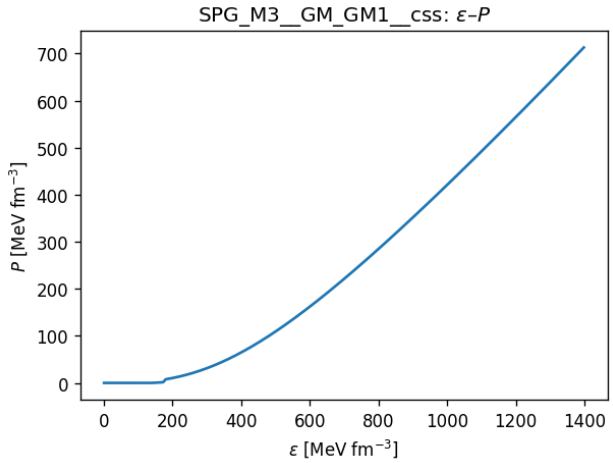

SPG_M1__APR_APR__css: ε–P
P [MeV fm^{-3}]
ε [MeV fm^{-3}]

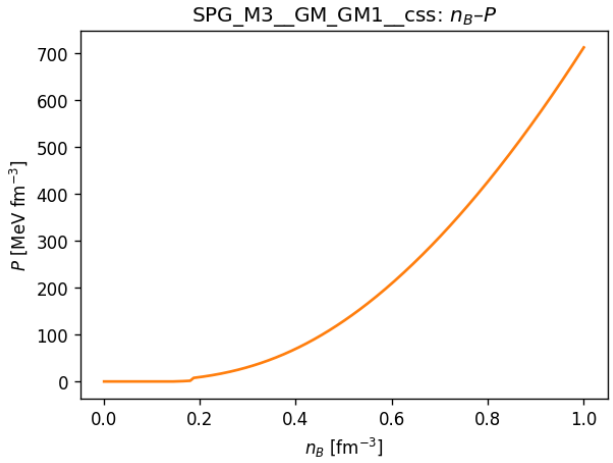

SPG_M1__APR_APR__css: n_B–P
P [MeV fm^{-3}]
n_B [fm^{-3}]

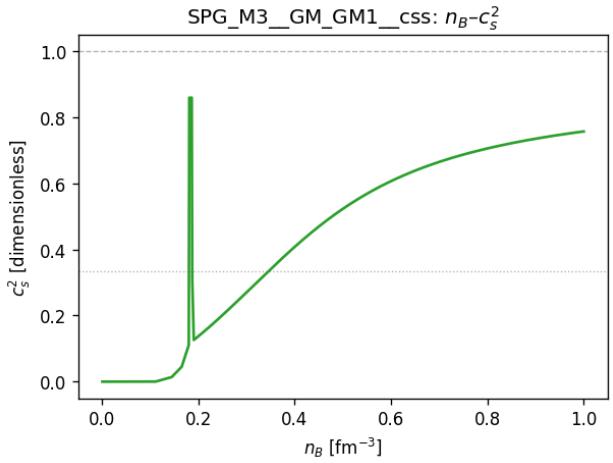

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c_s^2 [dimensionless]
n_B [fm^{-3}]

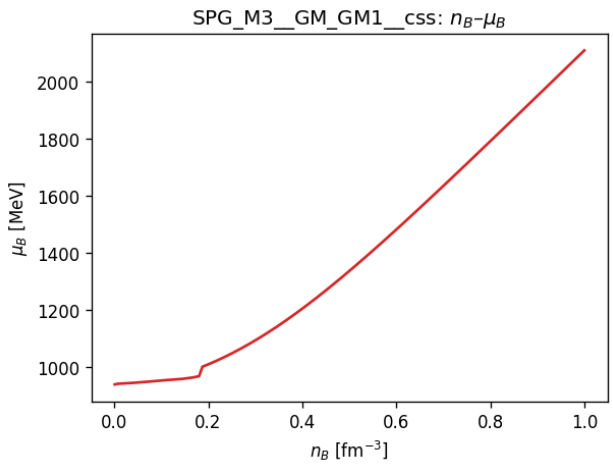

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μ_B [MeV]
n_B [fm^{-3}]

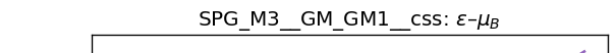

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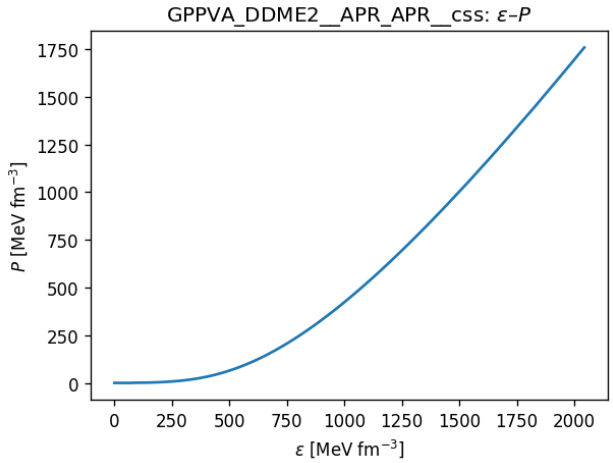

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P [MeV fm^{-3}]
ε [MeV fm^{-3}]

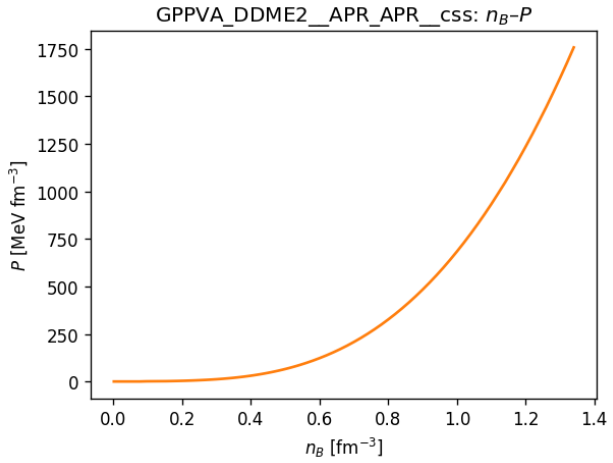

SPG_M3__GM_GM1__css: n_B–P
P [MeV fm^{-3}]
n_B [fm^{-3}]

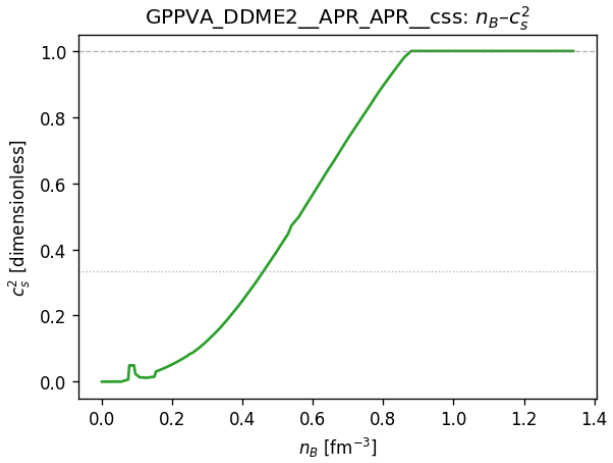

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c_s^2 [dimensionless]
n_B [fm^{-3}]

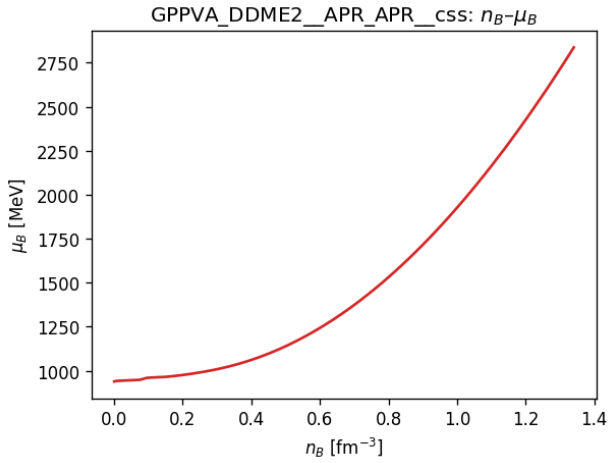

SPG_M3__GM_GM1__css: n_B–μ_B
μ_B [MeV]
n_B [fm^{-3}]

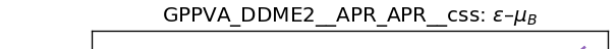

SPG_M3__GM_GM1__css: ε–μ_B


GPPVA_DDME2__APR_APR__css: ε–P
P [MeV fm^{-3}]
ε [MeV fm^{-3}]


GPPVA_DDME2__APR_APR__css: n_B–P
P [MeV fm^{-3}]
n_B [fm^{-3}]


GPPVA_DDME2__APR_APR__css: n_B–c_s^2
c_s^2 [dimensionless]
n_B [fm^{-3}]


GPPVA_DDME2__APR_APR__css: n_B–μ_B
μ_B [MeV]
n_B [fm^{-3}]


GPPVA_DDME2__APR_APR__css: ε–μ_B